

To Intervene or not to Intervene: Detecting and Examining Facilitator Tendencies in Online Discussions

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Abstract

The question of how and when to intervene in online discussions—whether through expert or natural facilitation—is a long-standing concern in the social sciences. This debate has gained new urgency with the advent of Large Language Models, which ostensibly make automated, large-scale intervention increasingly feasible. In this study, we examine human and Large Language Model tendencies of intervention, with a specific focus on positive and negative reinforcement. We then use transformers to [DETAILS TO BE ADDED]. Our results show [DETAILS TO BE ADDED].

address url <https://aclanthology.org/D19-1481> and 5 is I'm sure it was not limited to Helpful or clarifying explanations Respectful disagreement Constructive feedback Polite empathetic or disrespectful language Personal attacks or insults Repeated interruptions or monopolizing the exchange Escalating conflict rather than resolving it Contributions that reduce the overall quality of interactions Our survey synthesizes ideas from Natural Language Processing ' ISBN 979-8-89176-251-0 ' doi 10.50.0

1 Introduction

Given the scale of social media networks and online spaces, facilitation needs to be automated to keep up with the amount of posted content (Small et al., 2023; Burton et al., 2024). So far, platforms have attempted to solve this issue by ignoring it, preferring automated flag-and-removal mechanisms (“content moderation”) (Korre et al., 2025), which are not sufficient (Cresci et al., 2022; Trujillo and Cresci, 2022). While recent work has undertaken the task of choosing *how* to facilitate (Chen et al., 2025; Schroeder et al., 2024; Park et al., 2012), no studies so far have focused on *when* to do so.¹

Deciding when to intervene in a discussion is a matter of great importance. Intervening too early or too frequently may lead to users expressing frustration at being constantly interrupted, while intervening too late may lead to a significant escalation and derailment of the discussion (Korre et al., 2025; Gao et al., 2025). Since

¹We reference studies based on “discussion escalation” in §2.1, which focus on a very narrow subset of facilitation tasks.

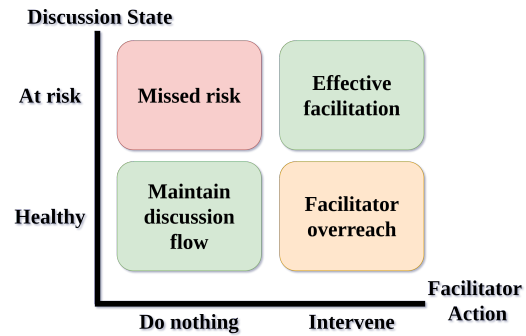


Figure 1: At any point in a discussion, the facilitator must decide whether to intervene or not. Not intervening when the situation calls for it may lead to the discussion worsening (e.g., topic derailment, escalation, missed participation). On the other hand, intervening when the discussion is already going well may lead to irritation among participants.

automation is needed, LLMs have been championed as the eventual solution to online facilitation (Korre et al., 2025; Small et al., 2023). However, previous work in LLM facilitation indicates that LLM facilitators are way too overeager to intervene, rendering them unusable as autonomous facilitative agents (Tsirmpas et al., 2026). In this study we focus on the following Research Question (RQ):

RQ. *Why do LLMs exhibit aberrant behavior in facilitative interventions?*

In order answer to the central research question we need to answer the following sub-questions:

What constitutes an intervention? There is currently no consensus on how to define a facilitative intervention. Additionally, current work on facilitation, including theoretical contributions, datasets, and annotations is heavily skewed towards specialized, oral forms of discussions, such as debates (Chen et al., 2025) and sharing personal stories (Schroeder et al., 2024). Thus, our work attempts to transfer knowledge from specialized, oral datasets from different domains, to the important field of (written) online facilitation. We find three common definitions implicitly used in the literature; escalation, professional facilitation, and facilitative comments (§2.1).

How do humans facilitate? Since datasets for discussion facilitation are generally few, small, and straddle radically different domains (e.g., debates, forums, deliberations), we gather all datasets relating to these three possible definitions of the term, and compile a standardized dataset for discussion facilitation with a unified data and label schema (§3). We then create and execute an annotation task designed to gauge human facilitative patterns by sampling 1226 discussion chunks from our dataset, and asking 10 expert annotators whether they would intervene as facilitators. We find that...

How does LLM facilitation differ? We repeat the annotation task with six open-source LLMs. Unlike humans...

Why does LLM facilitation differ? <Discussion section>

How can we fix it? <If LLMs are seeing something humans don't (they are proactive rather than reactive), then perhaps humans should start learning by their example>. Otherwise, we rely on professional facilitation datasets to transform the subjective task of facilitative interventions, into a classification problem. Using smaller, Deep Learning (DL)-based models, we can restrain LLMs to only facilitate when truly necessary (§5). <Mention LLM-as-a-judge models for comparison>.

2 Related Work

2.1 What are facilitative interventions?

Facilitation (or conversational moderation) refers to the active participation of people in order to improve discussions (Korre et al., 2025). Relevant work assumes that facilitation is conducted by specially trained staff (Schroeder et al., 2024; Park et al., 2012), users appointed by the community (Schaffner et al., 2024), or specialized professionals (Chen et al., 2025). We identify three main definitions for what a facilitative intervention is:

Definition 1 (Escalation). *A facilitator needs to intervene at the precise point in the discussion at which, without intervention, the discussion would derail or escalate.*

Escalation is usually measured as the point where the participants request arbitration (e.g., escalate the matter to a site moderator in Wikipedia (De Kock and Vlachos, 2021; Chang and Danescu-Niculescu-Mizil, 2019)). This definition is quite appealing for Natural Language Processing (NLP) researchers as the labels are easy to acquire since a ground truth exists (i.e., “escalation” can be defined in terms of moderator and user actions specific to each platform), and (2) the existence of a single, largely unambiguous ground-truth is ideal for the training and evaluation of any Machine Learning (ML)-based classifier. However, it is also inherently restrictive, as a facilitator has multiple duties that do not

directly relate to preventing personal attacks, or rhetorical escalation (Korre et al., 2025). Furthermore, the datasets require that the platforms contain an unambiguous signal that a discussion has escalated, which is not often the case (Table 1). Thus, this definition is not considered in this study, but is included in our dataset (§3).

Definition 2 (Professional facilitation). *An intervention should occur at the point where a professional facilitator would have intervened.*

Recent social science work insists that facilitators help discussions in many ways other than escalation prevention; such as summarizing discussions (Small et al., 2023), managing participation (Chen et al., 2025; Schroeder et al., 2024), and providing important information for new members (Park et al., 2012). It is generally accepted that all comments made by a facilitator should be considered facilitative interventions; for instance, all such comments are assumed to follow some social strategy (Chen et al., 2025; Schroeder et al., 2024). These authors thus implicitly or explicitly use the definition above to distinguish between facilitative and user comments in their work.

Defining facilitative comments this way is simple and intuitive, but can only be used in domains where (1) professional facilitators exist and (2) they can be identified. These conditions exclude the vast majority of online discussions, as very few web-crawled datasets include such information. This can be seen clearly in Table 1, where the extremely large WikiConv dataset (Hua et al., 2018) can not be used for analyzing facilitation using this definition, since there is no explicit information on which user is a facilitator (in this case, a Wikipedia moderator).

Definition 3 (Human judgment). *Facilitation should only occur when most people (annotators) agree that it is necessary.*

While some may argue that professional facilitators have more accurate judgments due to expertise, in reality most facilitation is handled by either volunteers elevated to the position by their community (Schaffner et al., 2024) or by the users themselves in an ad-hoc fashion (Falk et al., 2024). Given that humans generally hold extensive social experience because of sustained social interactions from birth, they innately understand concepts such as when to speak in a discussion (Nonomura and Mori, 2025). Thus, they are a good baseline for machine-enabled facilitation. In this case, this task is no longer about detecting or predicting whether a facilitator would speak, but rather the much more difficult and subjective task of whether the comment is facilitative in nature.

2.2 Spoken vs online facilitation

There currently exists a rift in facilitation research; while some previous work concerns itself with forums and online discussions (Park et al., 2012), others

are exclusively focused on live, real-life, oral discussions. (Schroeder et al., 2024; Chen et al., 2025). The presence of these two sub-modalities necessitates separate experiments for either of them, since they represent different kinds of data. Posts and comments for instance are self-contained, while transcripts regularly feature interruptions, participants talking in parallel (see App. ??), and linguistic features that are necessary for coordination (such as the dash character used for signaling an interruption). They may also feature fundamentally different kinds of facilitation interventions, as hinted by stark differences in proposed facilitation schemas between online forums (Park et al., 2012), and real-life discussions (Schroeder et al., 2024; Chen et al., 2025). It is worth noting that in our case, all written discussions are asynchronous (the participants may join and leave at any time), while all spoken discussions are synchronous (the participants are all active at the same time).

2.3 How to determine whether to facilitate

2.3.1 Don't

Dimitris. What most platforms do is to use flagging and reports to decide when to intervene. Mention Grok as a facilitative agent that works by invocation only. Briefly talk about advantages and disadvantages of current approach.

2.3.2 Using humans

Recent research states that a fully-automated facilitation is still out of our reach as LLMs tend to be very intruding and repetitive, and fixate on the moderation side of facilitation (Tsirmpas et al., 2026; Korre et al., 2025). Human facilitators, on the other hand, can step in the conversation with a more empathetic outlook, take into deep consideration the context of the conversation, while also being more sensitive to the differences and causes of certain social behaviors.

Disadvantages However, recruiting human facilitators, which also indicates a sort of expertise, is not a scalable process. In terms of LLM and human annotation alignment, (Movva et al., 2024) focusing on online safety contexts, show that LLMs are quite unidimensional and larger datasets are needed to understand if LLMs can adequately represent different demographic groups. This can also be translated in the context of facilitation, when LLMs *get triggered* only when toxicity appears in the text, disregarding the encouragement of prosocial behavior.

2.3.3 Using models

While some recent work (Schroeder et al., 2024; Chen et al., 2025) uses LLM-as-a-judge models to annotate data in the field of facilitation, the use of such models may not be an obvious choice for the task of facilitation prediction. It has been demonstrated that LLM facilitators can not decide when they should intervene irrespective of instructions and model family or size—instead, since these models are trained as chatbot assistants, they constantly attempt to speak (Tsirmpas et al.,

2026).¹⁷ On the other hand, the use of traditional ML and DL classifiers is often used in the context of escalation prediction (Chang and Danescu-Niculescu-Mizil, 2019; Yuan and Singh, 2023), and facilitative tactics detection (Chen et al., 2025; Schroeder et al., 2024).

3 Dataset

Since there is currently no dataset that explicitly concerns itself with facilitation interventions, we construct the “Prosocial and Effective Facilitation in Konversations (PEFK)” dataset, a standardized multi-domain compilation of the facilitation-related datasets listed in Korre et al. (2025) (see Table 1). PEFK contains data from both online forums and spoken (debates, radio, and deliberative) discussions, and standardizes discussion content, metadata, metrics, and labeling schemes according to a simple, unified schema, making it suitable as a benchmark on facilitator interventions.² More information on preprocessing steps and dataset schema can be found in App. B.

PEFK is designed to facilitate exploration and analysis of facilitative interventions under all three definitions described in §2.1. Specifically, we use a subset of the data that includes professional facilitation annotations to train our models according to Definition 2 (App. C.2.1). Given the extremely large size of the dataset, following human judgment to mark facilitative comments (Definition 3) would be prohibitively expensive. Thus, we annotate part of the dataset under a different annotation schema, creating a new evaluation set using expert annotators instead of professional facilitators according to Definition 3 (§4). Finally, our dataset can also be used for analyzing facilitation through the lens of escalation and toxicity, as argued in Definition 1, although this information is not utilized in this study (see §2.1).

4 Can humans and LLMs agree when to intervene?

4.1 Facilitative annotation

Labels Positive and negative reinforcement here should be understood as insights that emerged from our discussions with annotators, rather than as predefined moderation tools. In particular, positive reinforcement has often been overlooked, even though our goal in annotation and evaluation was not limited to moderation alone. Instead, we aimed to capture a broader range of interaction dynamics, including how constructive or supportive responses can be recognized and encouraged, not just how harmful content is flagged or constrained. At the same time, our analysis operates at a higher level, even though the annotators were extensively trained on established facilitation taxonomies (see App. D.1).

Data We randomly split multi-party discussions from the PEFK dataset into small chunks containing a set

²Specifically, interventions following the assumption of professional facilitators (Definition 2).

Name	#Comments	#Discussions	Domain	Format	Fac.
WikiDisputes (2021)	96,320	8,727	Forum	Text	×
WikiTactics (2022)	3,850	213	Forum	Text	✓
WikiConv (2018)	17,806,373	4,486,983	Forum	Text	×
Conversations Gone Awry / CMV II (2019)	40,607	6,691	Forum	Text	×
CeRI (2012)	3,700	132	Forum	Text	✓
User Moderation (UMOD) (2024)	2,000	1,000	Forum	Text	✓
Intelligence Squared 2 (IQ2) (2016)	34,245	108	Debate	Oral	✓
Why How Who (WHoW) (2025)	25,542	178	Radio / TV	Oral	✓
Fora (2024)	39,438	262	Deliberation	Oral	✓

Table 1: The datasets used in PEFK with accompanying post-processing statistics. We include a large variety of domains, and two different sources; text-only datasets, and transcripts from oral discussions. All datasets include multi-participant discussions, besides UMOD, which contains comment pairs. The **Fac.** column denotes whether information on which user is a facilitator is provided.

amount of comments (6 comments per discussion), and select a random sample for annotation (1224 discussions). This design aims to replicate a known limitation in online asynchronous facilitation scenarios; the facilitator is expected to decide on a course of action considering only the last parts of a discussion. Indeed, moderators tend to focus on the last-introduced comments in a discussion (Hewitt, 2003; Park et al., 2016; Kumpel and Unkel, 2021). This exact schema is also used in a crowdsourced moderation platform (Lampe et al., 2014). While this justification does not apply directly to spoken discussions, there is evidence to suggest that the most recent comments in a discussion are given relatively larger importance by the participants (Crano, 1977; Cockburn et al., 2017).

Annotators We recruited 10 expert annotators. All annotators had extensive experience with NLP annotation tasks from sentiment analysis to facilitation type categorization. The gender distribution was relatively balanced with 4 males and 7 females, while their ages had a median of 30 years. Before the commencement of the annotation process, the annotators completed an agreement form in compliance with GDPR (European Parliament and Council of the European Union) ethical requirements. They were compensated in accordance with national legislation.

Annotation Procedure We select a total of 1212 random discussions from all datasets that include information on facilitator interventions (Table 1) with a uniform distribution, and select a random section with length 6 comments or less from each.

Due to the lack of a unified and widely agreed-upon taxonomy of facilitation interventions (Korre et al., 2025), we focus on a single, broadly applicable dimension: *reinforcement*. Specifically, we distinguish between *positive reinforcement*, which aims to encourage or reward constructive behavior (e.g., expressing appreciation or affirming contributions), and *negative reinforcement*, which aims to discourage undesirable behavior or guide participants back to productive discourse (e.g., addressing incivility or off-topic comments). Our dichotomy is not intended as a theoretical replacement

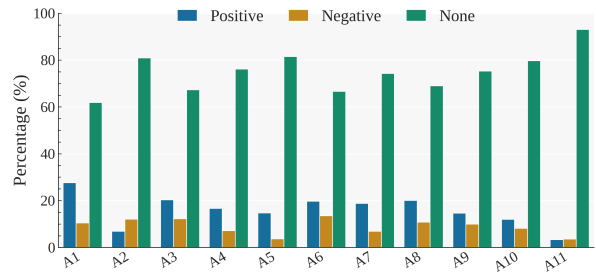


Figure 2: Distribution of all classes per annotator.

Category	Humans	Models
Positive	0.50	0.50
Negative	0.58	0.56
No-Reinforcement	0.47	0.46

Table 2: Inter-Annotator Agreement (IAA) (Fleiss κ) comparison within the human annotator group, model group, and both humans and models.

for these frameworks; rather, it serves as a practical abstraction that highlights the presence of facilitative intent in cases where annotators might otherwise overlook it. There is some theoretical grounding justifying this approach, described in App. D.3.

The task required the annotators to label instances from PEFK by determining whether, at any point within a specific dialogue ‘chunk’, reinforcement was needed, as well as what kind. We adopt an ordinal scale approach, where the annotators rate each dialogue for either type of reinforcement (positive or negative), as well as the possibility of no reinforcement using a scale from 1 to 5. The full annotation guidelines can be found in App. D.

4.2 Results

Apart from some *negligible* exceptions, human annotators very cautious when it comes to intervening, with most often selecting to not intervene (no reinforcement). When it comes to whether they choose positive or negative reinforcement, the discrepancy is not that great, with positive reinforcement being selected slightly more



Figure 3: Pairwise Cohen’s κ per class.

often. However, it must be noted (we need to connect this with the previous results), that despite the slightly higher frequency of positive reinforcement, they agree more on where intervene with negative reinforcement. We also run a small scale IAA analysis looking at the oral datasets (debates, and ..) in comparison to the written datasets (forums) and the difference in IAA is minimal with 30% and 33% Fleiss κ respectively.

Figure 6 shows the proportion of comments for which at least a given percentage of annotators agree that an intervention should occur. At the lower end (around 20–40% agreement), a large proportion of comments qualify. This suggests that many comments trigger at least some annotator belief that intervention is warranted, but that belief is not widely shared. As the agreement threshold increases (50–70%), the proportion drops substantially. This indicates moderate disagreement across annotators, with fewer cases reaching majority consensus. At high agreement levels (80–100%), only a small fraction of comments remain. This implies that strong consensus is rare, and only a limited subset of comments are clearly perceived as requiring intervention by most annotators.

4.3 Could we use LLMs instead of annotators?

The setup used by the authors simply prompted LLMs to intervene “when absolutely necessary”. Unlike their work, we explicitly ask the models whether they would have intervened without them engaging with the discussion, or explicitly having a facilitator role. We initially hypothesized that this setup would lead to the models being less prone to intervening, since they are no longer instructed to make the discussion better as a facilitator.³ However, the rates of interventions seem to be the same as the ones reported by the authors, suggesting that the *over-eagerness to intervene is unlikely to be solvable by simply using prompting*. Therefore, we believe that out-of-the-box LLMs can not be used for facilitation.

³The fact that LLMs are biased to always respond as chatbot assistants was hypothesized by the original authors to be the main reason this phenomenon occurs.

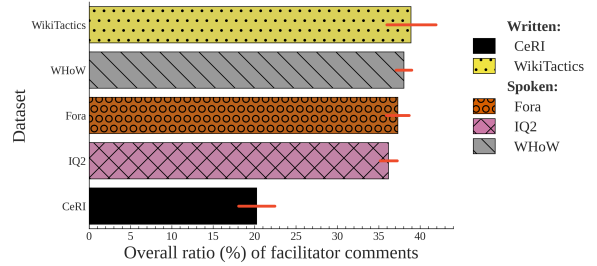


Figure 4: Percentage of moderated comments for each of the sub-datasets of PEFK. Bars denote 95% confidence intervals. Despite the different kinds of facilitators in online platforms and various real-life-discussion domains, the overall rate of facilitation remains similar across datasets.

5 Can we predict when a professional facilitator would intervene instead?

As discussed in §2.1, facilitation interventions can be defined in two ways; whether a comment is facilitative in nature (Definition 3), or whether a dedicated facilitator would participate at a given point in time (Definition 2). In this section we investigate whether we can use the few and limited available datasets to answer the latter definition.

Experimental setup We transform the task into a classification problem, where we are looking for the last comment made by a non-facilitator before a facilitator spoke. We train two ModernBERT (Warner et al., 2025) classifiers on the spoken and written datasets respectively.⁴ using the last comment before a facilitator intervention as the target label. We redact participant names, provide the 3 last comments as context, and use a 60-20-20 split keeping the same splits for all comparisons. We use a single Quadro RTX 6000 GPU. Training the model on the written datasets requires a few hours of computational time, while the spoken datasets require up to 32 hours, due to the size of the training data. See App. C.2.1 for more details concerning the training

⁴We also run an ablation study where a classifier is trained on all datasets at once (App. C.2.3), but performance suffers.

Dataset	Type	Pr.	Rec.	F1	Sup.
CeRI	W	0.320	0.490	0.387	192
WikiTactics	W	0.407	0.717	0.520	258
Fora	S	0.409	0.631	0.497	2764
IQ2	S	0.399	0.634	0.490	2393
WHoW	S	0.418	0.626	0.502	1816

Table 3: Classifier performance by dataset on the test-set and format (W=written, S=spoken). We evaluate precision, recall and binary-F1, and support (number of test examples).

setup.

Results Table 3 shows the results for the facilitator intervention task. We see that our DL classifiers are capable of modest performance in this task, scoring generally high recall, but lower precision scores across all datasets, despite the relatively small size of the training datasets. High recall in this case indicates that our classifiers can generally detect when a facilitator would intervene, while lower precision suggests that they would intervene in cases where a facilitator did not intervene.⁵:

Can we do better? We should note that there is an inherent ceiling to the performance of these classifiers, due to noise in our labeling scheme. Specifically, many comments made by a facilitator may not be facilitative in nature, while some non-facilitator comments are (Falk et al., 2024). Furthermore, many comments may be facilitative but not possible to distinguish from context alone—for instance phrases such as “Yes” and “Thank you” which are predominant in spoken discussions (Fig. 7) may be both used for social coordination (non-facilitative) and managing participation (facilitative). There is currently no consensus on how these phrases should be categorized, with relevant studies (Schroeder et al., 2024; Chen et al., 2025; Park et al., 2012) deferring to whether the speaker is a facilitator.

Indeed, we show that the DL based classifiers do not score reliably on the task of *detecting* facilitator comments either (App. C.2.4), suggesting that these models are limited by the difficulty in distinguishing between which user is a facilitator. Thus, significant improvements in performance can only be realized with the inclusion of larger professional facilitation datasets, whose interventions can be *reliably defined and distinguished* from non-facilitative comments.

Finding 1. *Facilitator interventions as defined by the literature can not be automatically distinguished from non-facilitator comments in many cases. Thus, we can not reliably detect many cases where a facilitator would have intervened.*

⁵The precision-recall tradeoff can be tuned at inference-time (without training) by adjusting the probability threshold over which the model determines that an intervention should happen; see App. C.2.2.

6 Conclusion

Humans are good but expensive. Our research is good. We wasted a good amount of money on this, please accept this.

7 Limitations

None, we are awesome.

8 Ethical Considerations

All experiments and data have passed through official ethical approval by an Ethics Committee, and are compatible with GDPR (European Parliament and Council of the European Union) and the EU AI Act (European Parliament and Council of the European Union, 2024). The annotators were compensated fairly with a monthly rate of 1000€ and signed informed consent forms before beginning annotation and were allowed to skip uncomfortable tasks at any point, with no required justification.

In order to help research in the field of facilitation, we openly release the code that constructs PEFK under a GPLv3 license. All datasets used in PEFK are obtained from the sources provided by the respective author. We do not distribute the corpus separately, and thus no additional data license is applicable. The licenses of the original distribution apply. The Fora dataset (Schroeder et al., 2024) is not publicly available; researchers should contact the authors of Fora for access.

We use exclusively open-source models, libraries and LLMs, in order to ensure reproducibility, protect user data, and comply with relevant regulations (see App. C.1).

References

- James Bisbee, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. 2024. [Synthetic replacements for human survey data? the perils of large language models](#). *Political Analysis*, 32(4):401–416.
- Jason W. Burton, Ezequiel Lopez-Lopez, Shahar Hechtlinger, Zoe Rahwan, Samuel Aeschbach, Michiel A. Bakker, Joshua A. Becker, Aleks Berditchevskaia, Julian Berger, Levin Brinkmann, Lucie Flek, Stefan M. Herzog, Saffron Huang, Sayash Kapoor, Arvind Narayanan, Anne-Marie Nussberger, Taha Yasseri, Pietro Nickl, Abdullah Almaatouq, and 9 others. 2024. [How large language models can reshape collective intelligence](#). *Nature Human Behaviour*, 8(9):1643–1655.
- Jonathan P. Chang and Cristian Danescu-Niculescu-Mizil. 2019. [Trouble on the horizon: Forecasting the derailment of online conversations as they develop](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 4743–4754, Hong Kong, China. Association for Computational Linguistics.

Ming-Bin Chen, Lea Frermann, and Jey Han Lau. 2025. WHoW: A cross-domain approach for analysing conversation moderation . In <i>Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)</i> , pages 2091–2126, Albuquerque, New Mexico. Association for Computational Linguistics.	pages 2070–2095, Vienna, Austria. Association for Computational Linguistics.	560 561
Andy Cockburn, Philip Quinn, and Carl Gutwin. 2017. The effects of interaction sequencing on user experience and preference. <i>International Journal of Human-Computer Studies</i> , 108:89–104.	Jim Hewitt. 2003. How habitual online practices affect the development of asynchronous discussion threads. <i>Journal of Educational computing research</i> , 28(1):31–45.	562 563 564 565
William D Crano. 1977. Primacy versus recency in retention of information and opinion change. <i>The Journal of Social Psychology</i> , 101(1):87–96.	Yiqing Hua, Cristian Danescu-Niculescu-Mizil, Dario Taraborelli, Nithum Thain, Jeffery Sorensen, and Lucas Dixon. 2018. Wikiconv: A corpus of the complete conversational history of a large online collaborative community . In <i>Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing</i> , pages 2818–2823, Brussels, Belgium. Association for Computational Linguistics.	566 567 568 569 570 571 572 573
Stefano Cresci, Amaury Trujillo, and Tiziano Fagni. 2022. Personalized interventions for online moderation . In <i>Proceedings of the 33rd ACM Conference on Hypertext and Social Media</i> , HT ’22, page 248–251, New York, NY, USA. Association for Computing Machinery.	Katerina Korre, Dimitris Tsirmpas, Nikos Gkoumas, Emma Cabalé, Danai Myrtzani, Theodoros Evgeniou, Ion Androutsopoulos, and John Pavlopoulos. 2025. Evaluation and facilitation of online discussions in the LLM era: A survey . In <i>Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing</i> , pages 24443–24462, Suzhou, China. Association for Computational Linguistics.	574 575 576 577 578 579 580 581
Christine De Kock, Tom Stafford, and Andreas Vlachos. 2022. How to disagree well: Investigating the dispute tactics used on wikipedia . In <i>Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing</i> , pages 3824–3837, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.	Anna Sophie Kümpel and Julian Unkel. 2021. (why) does comment presentation order matter for the effects of user comments? assessing the role of the availability heuristic and the bandwagon heuristic. <i>Communication Research Reports</i> , 38(4):217–228.	582 583 584 585 586
Christine De Kock and Andreas Vlachos. 2021. I beg to differ: A study of constructive disagreement in online conversations . In <i>Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume</i> , pages 2017–2027, Online. Association for Computational Linguistics.	Cliff Lampe, Paul Zube, Jusil Lee, Chul Hyun Park, and Erik Johnston. 2014. Crowdsourcing civility: A natural experiment examining the effects of distributed moderation in online forums. <i>Government Information Quarterly</i> , 31(2):317–326.	587 588 589 590 591
European Parliament and Council of the European Union. Regulation (EU) 2016/679 of the European Parliament and of the Council .	Rajiv Movva, Pang Wei Koh, and Emma Pierson. 2024. Annotation alignment: Comparing LLM and human annotations of conversational safety . In <i>Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing</i> , pages 9048–9062, Miami, Florida, USA. Association for Computational Linguistics.	592 593 594 595 596 597 598
European Parliament and Council of the European Union. 2024. Regulation (eu) 2024/1689 of the european parliament and of the council of 13 june 2024 laying down harmonised rules on artificial intelligence (artificial intelligence act) . https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng . Official Journal of the European Union, L 2024/1689.	Ryota Nonomura and Hiroki Mori. 2025. Who speaks next? multi-party ai discussion leveraging the systematics of turn-taking in murder mystery games . <i>Frontiers in Artificial Intelligence</i> , Volume 8 - 2025.	599 600 601 602
Neele Falk, Eva Vecchi, Iman Jundi, and Gabriella Lapesa. 2024. Moderation in the wild: Investigating user-driven moderation in online discussions . In <i>Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)</i> , pages 992–1013, St. Julian’s, Malta. Association for Computational Linguistics.	Deokgun Park, Simranjit Sachar, Nicholas Diakopoulos, and Niklas Elmqvist. 2016. Supporting comment moderators in identifying high quality online news comments. In <i>Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems</i> , pages 1114–1125.	603 604 605 606 607 608
Rena Gao, Ming-Bin Chen, Lea Frermann, and Jey Han Lau. 2025. Moderation matters: Measuring conversational moderation impact in English as a second language group discussion . In <i>Findings of the Association for Computational Linguistics: ACL 2025</i> ,	Joonsuk Park, Sally Klingel, Claire Cardie, Mary Newhart, Cynthia Farina, and Joan-Josep Vallbé. 2012. Facilitative moderation for online participation in erulemaking . In <i>Proceedings of the 13th Annual International Conference on Digital Government Research</i> , dg.o ’12, page 173–182, New York, NY, USA. Association for Computing Machinery.	609 610 611 612 613 614

Perspective API. 2026. [Perspective api](#). Accessed: 2026-02-24.

Brennan Schaffner, Arjun Nitin Bhagoji, Siyuan Cheng, Jacqueline Mei, Jay L Shen, Grace Wang, Marshini Chetty, Nick Feamster, Genevieve Lakier, and Chenhao Tan. 2024. "community guidelines make this the best party on the internet": An in-depth study of online platforms' content moderation policies. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, CHI '24, New York, NY, USA. Association for Computing Machinery.

Hope Schroeder, Deb Roy, and Jad Kabbara. 2024. [Fora: A corpus and framework for the study of facilitated dialogue](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13985–14001, Bangkok, Thailand. Association for Computational Linguistics.

Christopher T. Small, Ivan Vendrov, Esin Durmus, Hadjar Homaei, Elizabeth Barry, Julien Cornebise, Ted Suzman, Deep Ganguli, and Colin Megill. 2023. [Opportunities and risks of llms for scalable deliberation with polis](#). *ArXiv*, abs/2306.11932.

Amaury Trujillo and Stefano Cresci. 2022. [Make reddit great again: Assessing community effects of moderation interventions on r/the_donald](#). *Proc. ACM Hum.-Comput. Interact.*, 6(CSCW2).

Dimitris Tsirmpas, Ion Androutsopoulos, and John Pavlopoulos. 2026. [Designing synthetic discussion generation systems: A case study for online facilitation](#). *Preprint*, arXiv:2503.16505.

Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, Griffin Thomas Adams, Jeremy Howard, and Iacopo Poli. 2025. [Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2526–2547, Vienna, Austria. Association for Computational Linguistics.

Jiaqing Yuan and Munindar P. Singh. 2023. [Conversation modeling to predict derailment](#). *Proceedings of the International AAAI Conference on Web and Social Media*, 17(1):926–935.

Justine Zhang, Ravi Kumar, Sujith Ravi, and Cristian Danescu-Niculescu-Mizil. 2016. [Conversational flow in Oxford-style debates](#). In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 136–141, San Diego, California. Association for Computational Linguistics.

A Acronyms

NLP Natural Language Processing

ML Machine Learning

RQ Research Question

PEFK Prosocial and Effective Facilitation in Konversations

DL Deep Learning

IAA Inter-Annotator Agreement

UMOD User Moderation

IQ2 Intelligence Squared

CeRI Cornell eRulemaking Initiative

WHoW Why-How-Who

B Dataset details

B.1 Preprocessing

General

- We exclude comments with no text.
- We exclude discussions with less than two distinct participants.
- We exclude discussions which are common between wiktactics and wikiconv as well as wikidisputes and wikiconv.
 - There may be duplicate discussions between wikidisputes and wiktactics, but we allow them since they feature complementary information.

Wikiconv

- The Wikiconv corpus does not contain information about which user is a moderator/facilitator. Therefore, all comments relating to Wikiconv are tagged as non-moderators.
- Additionally, we follow the instructions of the original researchers ([Hua et al., 2018](#)), and select only discussions which have at least two comments by different users.
 - Wikipedia (thankfully) does not track users who log in with only an IP address (in the original dataset, their user_id is always set to 0 and their username is of the form 211.111.111.xxx. We consider each such username to be a separate user.
 - Due to the size of the dataset, we have to partially load it during preprocessing. Thus, there is a small chance every 100,000 records that a discussion is marked as a false negative and a part of it gets discarded.
 - We only include English comments in the final dataset. We use a small, efficient library (py3langid) for language recognition, due to the large size of Wikiconv. Non-English comments are discarded before selecting valid discussions (see point above).

Name	Type	Description
conv_id	string	The discussion’s ID. Comments under the same discussion refer to the same discussion ID.
message_id	string	The message’s (comment’s) unique ID.
reply_to	string	The ID of the comment which the current comment responds to. nan if the comment does not respond to another comment (e.g., it’s the Original Post (OP)).
user	string	Username or hash of the user that posted the comment
is_moderator	bool	Whether the user is a facilitator.
moderation_supported	bool	True if the moderation labels are directly computed from the original dataset
escalated	bool	A discussion-level measure denoting discussions which have been de-railed
escalation_supported	bool	True if the escalation labels are directly computed from the original dataset
text	string	The contents of the comment
dataset	string	The dataset from which this comment originated from
notes	JSON	A dictionary holding notable dataset-specific information
toxicity	float	The “toxicity” score given to the comment by the Perspective API
severe_toxicity	float	The “severe toxicity” score given to the comment by the Perspective API

Table 4: The full set of columns used for each comment in the [PEFK](#) dataset.

Wikitactics

- We infer facilitative actions by whether the comment belongs in any of the following categories:
 - Asking questions
 - Coordinating edits
 - Providing clarification
 - Suggesting a compromise
 - Contextualisation
- The above tactics are a subset of the Coordinative labels used in the WikiTactics paper. They were selected because they are not used necessarily on 1-1 discussions; they could reasonably be applied by third-party participants. Contrast them with other Coordinative labels such as “Conceding/recanting” and “I don’t know”.

Wikidisputes

- Since only 0.03% of the comments in the dataset are made by moderators, we mark the dataset as not supporting moderation.

UMOD

- Facilitative actions are marked as a gradient from 0 (no facilitation) to 1 (full facilitation). We adopt a threshold of 0.75 to consider an action as facilitative, with more than 50% annotator agreement (measured as entropy in the original dataset).

CMV-AWRY2

- We mark a discussion as escalated when the derailment value (from the official dataset) is in the 60th upper percentile.
- We remove deleted (“[deleted]”) comments.

B.2 Schema

Table 4 details the information available for each comment in the dataset. Some information, such as whether the person talking is a moderator / facilitator, is available only on some sub-datasets. Also note that the `is_moderator` column may hold different definitions of what we consider as moderator / facilitator. UMOD for instance considers facilitative comments made by users, while others such as Fora consider a comment facilitative if it was made by a dedicated facilitator.

B.3 Exploratory Data Analysis

We explore some initial frequencies with regard to facilitator comments in [PEFK](#), comparing its spoken vs. written subsets, as well as the agreement and facilitator intervention frequencies in the re-annotated subset. **We examine our annotated dataset in terms of class distributions and inter-annotator agreement to identify any notable patterns or outliers.** Figure 2 presents the distribution of all classes across annotators. Discrepancies are evident: A11 appears particularly cautious, rarely assigning any type of intervention, while A2 is the only annotator who frequently uses negative reinforcement. Annotator A1 proved problematic, as they labeled the majority of instances as None, making them an outlier; consequently, A1 is excluded from subsequent analyses. A11 is also considered an outlier according to Figure 5

Spoken vs. written datasets in PEFK Figure 7 shows some of the structural differences of spoken and written datasets. Spoken datasets feature a significantly larger number of comments per discussion, since instances where the participants talk over each other, or interrupt the other are transcribed as separate comments.

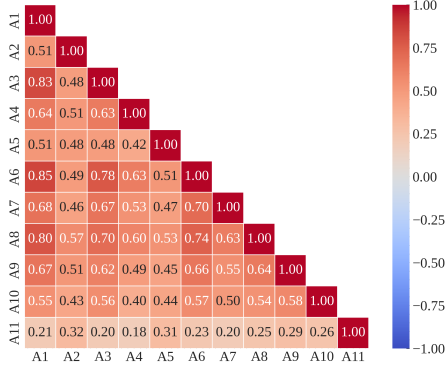


Figure 5: Heatmap featuring the pairwise Cohen’s κ of the annotators.

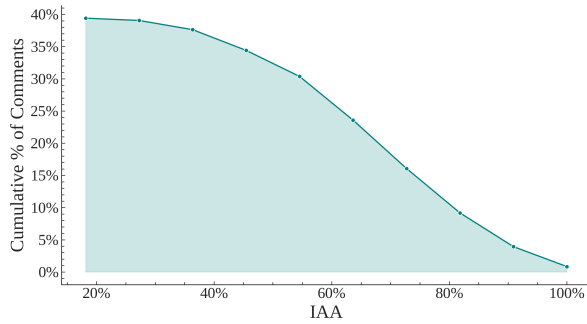


Figure 6: Number of comments plotted to the number of annotators that believe an intervention should occur.

Additionally, spoken datasets feature a large amount of comments with very few words, since a lot of utterances are short but necessary for social co-ordination (e.g., “Thank you”, “Yes”, “I agree”). Figure 4 shows the percentage of comments made by facilitators for each of our datasets. In almost all datasets, a third of the comments are made by facilitators.⁶

Vocabulary discrepancies in PEFK datasets. We also filter the data to include only messages written by moderators, then cleaned the text by removing URLs, punctuation, and other noise. After preprocessing, we used a TF-IDF vectorization approach with n-grams (bi-grams and trigrams) to extract phrases. Crucially, the TF-IDF model was trained on all datasets combined, so that common, generic phrases were downweighted while phrases that are more distinctive to each dataset were emphasized. We then computed the average TF-IDF score of each phrase within each dataset and selected the top-ranking ones as representative expressions of moderator language in that context. The results can be found in Figure 8.

The most important phrases in Cornell eRulemaking Initiative (CeRI) suggest a formal, policy-oriented environment with an emphasis on structured participation and regulatory discussion (e.g., references to “pro-

⁶We do not include the UMOD dataset, since there is no established ground-truth (see App. B.1).

posed rule” and “welcome regulations”). In *fora*, the language is more conversational and facilitative, with phrases such as “great thank”, “feel free”, and “awesome thank” indicating a more informal moderation style. The Intelligence Squared (IQ2) dataset reflects the debate setting, with phrases like “ladies gentlemen” and “arguing motion”. In User Moderation (UMOD), moderators engage more in argumentation and clarification, as seen in phrases like “change view”, “missing point” reminiscing more facilitative settings and suggesting a focus on reasoning and discussion quality. The Why-How-Who (WHoW) dataset appears similar to moderated discussions or panels, with phrases like “question sir” and “answer question” indicating turn-taking and audience interaction. Finally, *WikiTactics* shows a highly procedural and rule-based style, with phrases such as “reliable sources”, “original research” and “talk page” reflecting the norms and policies of collaborative editing environments like Wikipedia.

C Models

C.1 LLMs

We use only open-source models, as proprietary models such as OpenAI’s GPT-5 yield much less reproducible results (Bisbee et al., 2024). Furthermore, the use of open-source models requires much fewer resources, enabling research on this task with a less demanding budget, as well as scalable deployment in real-world settings (Tsirmpas et al., 2026). It is also worth noting that since a part of our dataset (Fora) is not licensed, we can not risk leaking the data contained within to proprietary models. Generally, the use of local models protects the personal data of the users contained in our datasets, ensures that our research can be replicated in domains where anonymity must be maintained, and renders our work compatible with existing legislation (European Parliament and Council of the European Union, 2024; European Parliament and Council of the European Union).

Table 5 shows details on the LLMs used for the annotation in this study. We specifically use 4-bit quantized versions from the unsloth huggingface repository.

Family	Version	#Params
OLMo	2 (0325)	32B
Qwen	2.5	32B
LLaMa	3.3	70B
OLMo	3	7B
Qwen	2.5	7B
LLaMa	3.1	8B

Table 5: LLMs used for annotation in this study.

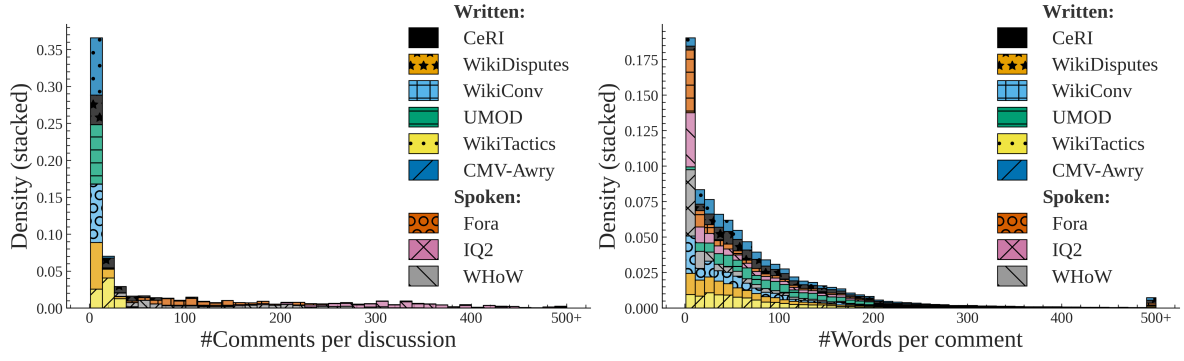


Figure 7: **Left:** Comments per discussion on the PEFK dataset. Note the significant increase in comments made in transcribed datasets such as Fora and IQ2. This deviation is explained by the fact that a single human utterance is regularly broken up to many comments in order to simulate interruptions or participants talking over each other. **Right:** Number of words per comment in all discussions in PEFK, grouped by original dataset. Comments across written datasets follow the same distribution, while spoken datasets feature a significant number of comments with very few words.

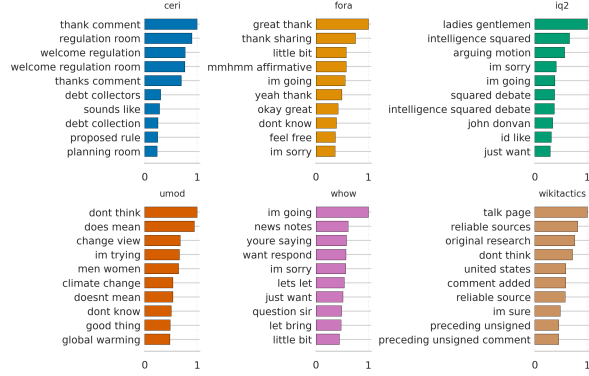


Figure 8: Most common moderator bigrams in PEFK per dataset. Function words are excluded.

C.2 Transformers

C.2.1 Training details

We construct our training and evaluation data from a unified corpus of threaded discussions, where each instance corresponds to a target comment augmented with its conversational context. For every target comment, we retrieve up to $K = 3$ preceding comments along the reply chain. Each comment is truncated to a maximum of 3000 characters to control sequence length. The input is formatted using XML-style tags, where context comments are wrapped with <CTX> and the target comment with <TGT>. The final sequence is formed by concatenating context comments in chronological order followed by the target comment. We consider binary classification tasks (e.g., is_moderator, should_intervene), with labels cast to floating point values.

We fine-tune a pretrained transformer encoder (ModernBERT-large) with an added classification head that outputs a single logit. The task is formulated as a single-label instance of multi-label classification, enabling the use of a sigmoid activation.

Training is conducted using the Hugging Face Trainer framework with several modifications. We optimize the model using binary cross-entropy with logits:

$$\mathcal{L} = \text{BCEWithLogitsLoss}(\mathbf{z}, \mathbf{y}; \text{pos_weight}),$$

where $\mathbf{z}$ are the predicted logits and $\mathbf{y}$ are the ground-truth labels. To account for class imbalance, the positive class is weighted as

$$\text{pos_weight} = \frac{N_{\text{negative}}}{N_{\text{positive}}}.$$

We optionally freeze the base transformer parameters and train only the classification head; this setting is enabled in our experiments to reduce computational cost and limit overfitting.

To improve computational efficiency, we employ a bucketed batching strategy that groups instances of similar length (approximated in characters) into batches of size 64. This reduces padding and improves throughput. Batches are shuffled at the bucket level at each epoch.

Training is performed for up to 80 epochs, with evaluation, logging, and checkpointing conducted once per epoch. The best model is selected based on validation loss. We apply early stopping with a warm-up phase: monitoring begins after one evaluation step, and training stops if validation loss does not improve by at least $\delta = 0.001$ for 6 consecutive evaluation steps. Model performance is evaluated using accuracy and binary F1 scores.

All experiments are implemented in PyTorch using the Hugging Face Transformers library. Data loading uses 4 worker processes, and GPU acceleration is utilized when available. We fix the random seed to 42 to ensure reproducibility.

C.2.2 Decision thresholds

An advantage of using DL-based classifiers instead of LLMs is that we can quantitatively alter the performance

of the classifier by tuning the thresholds required for a comment to trigger an intervention. By making the model more conservative (increasing the threshold), we could hope that we can avoid unnecessary interventions, while making it less conservative (decreasing the threshold) we could make it more proactive.

Fig. 9 shows that the classifier has an inherently low precision score which is largely invariant to the selected threshold. On the other hand, recall falls off rapidly, which incentivises us to use a more liberal approach.

C.2.3 One model to rule them all?

Since comments in the written and spoken datasets are fundamentally different (§2.2), we would be tempted to train different models for each. This decision is not obvious however, since it is likely that the model can internally distinguish between spoken and written speech. If that occurs, the differences between the texts should not negatively affect the model; in fact, training the model on all datasets could allow it to transfer knowledge between the two text modalities. Table 6 shows the performance of the same model when trained only on spoken and written datasets, compared to when trained on the all datasets at once. We find that the models perform better when trained on different splits.

Split	Dataset	Precision	Recall	F1
Written	CeRI	0.320	0.490	0.387
	Wikitactics	0.407	0.717	0.520
Spoken	Fora	0.409	0.631	0.497
	IQ2	0.399	0.634	0.490
	WHoW	0.418	0.626	0.502
All	CeRI	0.292	0.172	0.216
	Fora	0.410	0.664	0.507
	IQ2	0.406	0.623	0.491
	WHoW	0.422	0.642	0.509
	Wikitactics	0.456	0.364	0.405

Table 6: Summary of classifier performance depending on the training dataset splits on the facilitator prediction task. We compare models trained on all available data with models trained separately on written and spoken datasets. Performance is generally similar.

C.2.4 Facilitator detection

We train two ModernBERT classifiers on the task of facilitator detection, using the same training setup and parameters (App. C.2.1), where they attempt to guess whether a comment was made by a professional facilitator. This objective is contrasted with the task of facilitator prediction presented in §5, where the models attempted to guess whether a professional facilitator would have intervened.

While the tasks are similar on a technical level (the labels for the prediction task are the labels for the detection task, shifted by one to the previous comment), the objectives should have vastly different difficulties.

Specifically, we expect the detection task to be much simpler, since the models can understand whether an intervention occurred by examining the text of the professional facilitator, which should be structurally different to an extent from most user comments.

Table 7 shows the performance of these two classifiers. Contrasting with Table 6, we note a notable increase in the performance of the detection over the prediction models, suggesting that the task is indeed easier. However, the performance of the detection classifiers remains relatively low, which reinforces our hypothesis that many comments by professional facilitators are not distinctly facilitative compared to comments made by users (Finding 7).

Split	Dataset	Precision	Recall	F1
Written	CeRI	0.571	0.763	0.654
	Wikitactics	0.509	0.780	0.616
Spoken	Fora	0.467	0.617	0.532
	IQ2	0.550	0.680	0.608
	WHoW	0.507	0.617	0.557

Table 7: Summary of classifier performance depending on the training dataset split (Written vs. Spoken) on the facilitator detection task. The classifiers perform better on this task, but overall performance remains mediocre.

D Human annotation procedure

D.1 Facilitative tactics

Before the main annotation task on facilitative intervention, all annotators were extensively trained on currently established facilitative tactics. This task aimed to familiarize the annotators with what constitutes a facilitative intervention, as well as how professional facilitators operate in real-life scenarios.

Specifically, we split the annotators into three groups, each of which was assigned a different taxonomy of facilitative interventions (Chen et al., 2025; Schroeder et al., 2024; Park et al., 2012). The annotators were tasked with classifying 830 facilitated comments from a sample of continuous discussions in PEFK. The guidelines and results of this annotation are not shown in this study, but they are available upon request.

D.2 Expert facilitators vs. expert annotators

In designing our data annotation procedure, we intentionally prioritized the recruitment of expert annotators over expert moderators. This distinction is pivotal to our research design: whereas expert moderators typically engage in post hoc intervention—addressing content only after its publication—expert annotators facilitate a more proactive analysis of the discourse as it unfolds. However, acknowledging the importance of moderation experience we also include a professional moderator into the workflow to provide a more just comparison. In other words, we operate under the premise that the

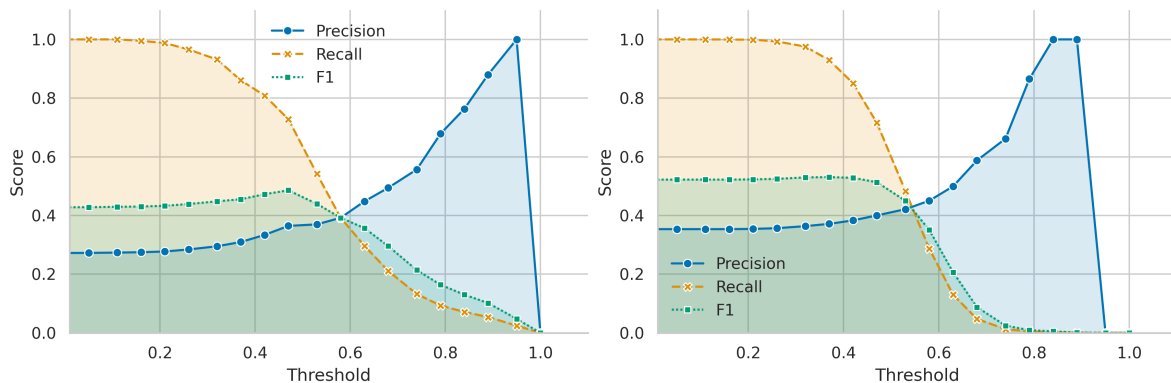


Figure 9: Classifier performance (**left**: written datasets, **right**: spoken datasets) when tuning the decision threshold on the test-set. The higher the threshold, the more conservative the classifier is encouraged to become when predicting a facilitator intervention.

majority of discussion facilitation and moderation is conducted organically by the participants themselves through user-driven flagging mechanisms or actual dialogue.

A11 does not exhibit the characteristics one would expect from a distinct “expert” annotator, but instead behaves as a highly balanced and representative member of the group. In contrast to other annotators who show clear directional biases—either favoring positive reinforcement (e.g., A1, A7, A8) or leaning negative (e.g., A2, A6)—A11 distributes judgments almost evenly between positive (28.80%) and negative (28.12%) labels, resulting in a near-zero difference (0.69 percentage points). At the same time, A11’s intervention rate (56.92%) is almost identical to the group average, indicating that it is neither more strict nor more lenient in deciding when to intervene. When considered alongside the moderate correlation scores with other annotators (mean 0.38), this pattern suggests that A11 does not consistently align more strongly with the group than others do, nor does it systematically deviate in a principled way. Instead, A11 appears to occupy the “center” of the annotator space: its behavior closely reflects the aggregate tendencies of the group without amplifying any particular bias. Consequently, the data do not support interpreting A11 as a clear expert or gold-standard annotator; rather, A11 is better characterized as a calibrated, balanced annotator whose decisions approximate the overall distribution of judgments in the dataset.

D.3 Positive vs. negative reinforcement

The distinction between positive/negative reinforcement is present to suggested facilitative tactics taxonomies (Table 1), including prior work such as Fora (Schroeder et al., 2024) WHoW (Chen et al., 2025), and CeRI (Park et al., 2012). We conservatively estimate whether the tactics described in each taxonomy correspond to our definitions of positive and negative reinforcement.

The facilitative taxonomy proposed in Fora does not include any tactics that could be unambiguously considered as negative reinforcement, which is expected,

since facilitators deployed in the relevant study were supposed to only act as coordinators. Notably, several tactics may simultaneously exhibit both positive and negative characteristics across all taxonomies, depending on context and the exact implementation used by the facilitator. We therefore use these taxonomies only as guidance during annotation, and refer the reader to the original works for a more comprehensive description and analysis of facilitation strategies.

E Annotation guidelines

E.1 Human guidelines

Annotation Guidelines: Type of Reinforcement Needed in Comment Threads

Purpose of the Task

These guidelines explain how to annotate comments in conversation threads using three parallel rating columns:

- Positive Reinforcement (1–5)
- Negative Reinforcement (1–5)
- No Reinforcement Needed (1–5)

Your job is to read each comment in its conversational context and assign a strength/certainty rating (1–5) for each category, reflecting how that category applies. All fields need to be filled and the categories are not necessarily mutually exclusive.

Even though your intervention will be delivered at the end of the thread, the reinforcement can refer to any point in the conversation, not just at the end. It can refer to something said earlier, not just the last comment.

You are not being asked whether that conversation correlates to positive or negative reinforcement, but rather whether you, personally, would have used positive or negative reinforcement at a given point. Keep in mind that too many interventions on the moderator’s part may result in an adverse reaction from the participants; no one wants to be constantly told what to say or how to say it.

Important Notes

- You must add the rationale behind your response in the rationale/comments field when:

Taxonomy & Tactic	Type
Fora	
Express appreciation	+
Express agreement or affirmation	+
Open invitation to participants	
Specific invitation to participant	
Model examples	
Follow-up question	
Make connections	
WHoW	
Informational	
Coordinative	-
Social	+
Probing	
Confronting	
Instruction	-
Interpretation	
Supplement	
CeRI	
Welcoming	+
Encouragement / appreciation	+
Maintaining civil discourse	-
Explaining why comment not removed	
Indicating irrelevant/off-point comments	-
Pointing to relevant information	
Pointing out effective commenting	+
Asking for more information/sources	
Asking for solutions/alternatives	
Encourage engagement with others	
Pose question to community	

Table 8: Facilitation taxonomies (bold) with associated tactics (normal text) and reinforcement type. +: positive reinforcement, -: negative reinforcement, empty if the tactic does not clearly correspond to any category.

Annotator	Correlation with A11
A6	0.492
A10	0.464
A2	0.432
A8	0.430
A1	0.396
A9	0.385
A4	0.383
A3	0.363
A12	0.301
A7	0.279
A5	0.249
Mean	0.380
Min	0.249
Max	0.492

Table 9: Correlation of Expert Moderators with Other Annotators

- your positive score > 3, or
- your negative score > 3, or
- you encounter a particular case (e.g., an instance which could fit both negative and positive reinforcement categories).
- If you encounter data that is malformed or noisy to the point that it renders comprehension difficult,

flag the instance using the malformed field.

What Each Label Means

Positive Reinforcement

Positive reinforcement is defined as an intervention whose purpose is to encourage a specific behavior. For example, a moderator may ask people to participate or ask a person to elaborate. Positive reinforcement is not related to positive speech; the moderator may be strict or even rude when attempting to encourage specific behaviors (e.g., “I will not tolerate any further discussion on this topic. It’s been decided and we’re moving on.”).

Rate this when a conversation demonstrates behavior that should be encouraged or reinforced in a synchronous exchange. Your task is not simply to identify positive behaviors, but to identify moments where reinforcing them would help sustain or build constructive conversational momentum.

This includes, but is not limited to:

- Helpful or clarifying explanations
- Respectful disagreement
- Constructive feedback
- Polite, empathetic, or supportive tone
- Efforts to de-escalate conflict
- Any contribution that meaningfully improves the quality or direction of the conversation

Negative Reinforcement

The purpose of negative reinforcement is to discourage unwanted or disruptive behaviors. This is correlated, but not necessarily closely related to threats or enforcement of disciplinary action. Negative reinforcement is not related to negative speech; the moderator may be kind and diplomatic when discouraging behaviors (e.g., “I’ve received multiple complaints about your behavior on this forum. Please refrain from making personal attacks and inflammatory comments.”).

Rate this when the comment reflects harmful, disruptive, or inappropriate behavior that should be discouraged, as well as cases where some participants dominate the conversation, or when the conversation starts to go off topic. Other examples include:

- Hostile, dismissive, or disrespectful language
- Personal attacks or insults
- Repeated interruptions or monopolizing the exchange
- Escalating conflict rather than resolving it
- Contributions that reduce the overall quality of the conversation

No Reinforcement Needed

Rate this when the comment is neutral, unremarkable, or does not affect the tone or direction of the conversation. This includes:

- Basic factual statements
- Clarifying questions
- Minimal or vague comments
- Harmless but non-notable contributions

How to Make Annotation Decisions

Look at Intent and Impact

Annotator	Negative (%)	No Reinforcement (%)	Positive (%)	Intervention Rate (%)	Pos - Neg (pp)
A1	27.61	31.99	40.40	68.01	12.78
A2	27.65	49.62	22.73	50.38	-4.92
A3	39.77	13.65	46.57	86.35	6.80
A4	23.82	47.59	28.59	52.41	4.77
A5	20.78	47.95	31.27	52.05	10.50
A6	24.73	53.11	22.16	46.89	-2.57
A7	19.84	46.21	33.95	53.79	14.11
A8	24.47	38.93	36.60	61.07	12.13
A9	25.52	43.51	30.97	56.49	5.45
A10	24.52	43.32	32.15	56.68	7.63
A11	28.12	43.08	28.80	56.92	0.69
A12	17.81	65.40	16.79	34.60	-1.01

Table 10: Behavioral Summary of Annotators (Percentages)

Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

Use the Context

Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context—and vice versa.

Judge Behavior, Not Opinions

The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

Remain Consistent

Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

E.2 LLM prompts

E.2.1 Yes/no intervention

You are an expert annotator specializing in online moderation and facilitation. You will be presented with a comment and the comments made before it. Choose whether you decide to intervene at this point in the discussion.

Key Rules for Annotating:

Look at Intent and Impact Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

Use the Context Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context—and vice versa.

Judge Behavior, Not Opinions The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

Remain Consistent Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

Output Respond whether you would intervene in this discussion as a facilitator on a scale of 1-5, where 1 is "I'm sure I don't need to intervene", and 5 is "I'm sure I need to intervene"

E.2.2 Positive vs. negative reinforcement

You are an expert facilitator. You will be presented with a discussion. Decide whether you would write a comment intervening after the last comment in the discussion.

The Core Rating Task

For every comment, you must write out a response of 1 (uncertain) to 5 (certain):

Positive Reinforcement

Defined as an intervention whose purpose is to encourage a specific behavior. For example, a moderator may ask people to participate, ask a person to elaborate etc.

Positive reinforcement is not related to positive speech; the moderator may be strict or even rude when attempting to encourage specific behaviors (e.g., "I will not tolerate any further discussion on this topic. It's been decided and we're moving on.").

Rate this when a conversation demonstrates behavior that should be encouraged or reinforced in a synchronous exchange. Your task is not simply to identify positive behaviors, but to identify moments where reinforcing them would help sustain or build constructive conversational momentum.

This includes, but is not limited to:

- Helpful or clarifying explanations
- Respectful disagreement
- Constructive feedback
- Polite, empathetic, or supportive tone
- Efforts to de-escalate conflict
- Any contribution that meaningfully improves the quality or direction of the conversation

Negative Reinforcement

The purpose of negative reinforcement is to discourage unwanted or disruptive behaviors. This is correlated, but not necessarily closely related to threats or enforcement of disciplinary action.

Negative reinforcement is not related to negative speech; the moderator may be kind and diplomatic when discouraging behaviors (e.g., "I've received multiple complaints about your behavior on this forum. Please refrain from making personal attacks and inflammatory comments.").

1211	Rate this when the comment reflects harmful, disruptive,	1268
1212	or inappropriate behavior that should be discouraged,	1269
1213	as well as cases where some participants dominate	1270
1214	the conversation, or when the conversation starts to go	1271
1215	off topic.	1272
1216	Other examples include:	1273
1217	• Hostile, dismissive, or disrespectful language	1274
1218	• Personal attacks or insults	1275
1219	• Repeated interruptions or monopolizing the exchange	1276
1220		1277
1221	• Escalating conflict rather than resolving it	1278
1222	• Contributions that reduce the overall quality of the	1279
1223	conversation	1280
1224	No Reinforcement	1281
1225	Rate this when the comment is neutral, unremarkable,	1282
1226	or does not affect the tone or direction of the conversation.	
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1228	This includes:	
1229	• Basic factual statements	
1230	• Clarifying questions	
1231	• Minimal or vague comments	
1232	• Harmless but non-notable contributions	
1233	Key Rules for Annotating	
1234	Look at Intent and Impact	
1235	Consider what the commenter seems to be trying to do,	
1236	and how their words might affect others in the thread.	
1237	Use the Context	
1238	Always check the surrounding messages. A comment	
1239	that looks fine on its own might escalate tension when	
1240	viewed in context—and vice versa.	
1241	Judge Behavior, Not Opinions	
1242	The label should never depend on whether you personally	
1243	agree with the comment. Focus on tone, clarity,	
1244	constructiveness, and how it influences the conversation.	
1245		
1246	Remain Consistent	
1247	Apply the same standards across all comments. Avoid	
1248	guessing or over-interpreting; use what is present in the	
1249	text.	
1250	Output	
1251	Respond whether you would intervene in this discussion	
1252	as a facilitator.	
1253	Always respond with the following format and nothing	
1254	else:	
1255	Positive: 1-5	
1256	Negative: 1-5	
1257	No Reinforcement: 1-5	
1258	E.2.3 Single-comment facilitation prediction	
1259	You are an expert moderator/facilitator. You will be	
1260	presented with a comment (TGT) and the comments	
1261	made before it (CTX, one for each comment in turn).	
1262	Choose whether you decide to intervene at this point in	
1263	the discussion.	
1264	Key Rules for Annotating:	
1265	Look at Intent and Impact Consider what the commenter	
1266	seems to be trying to do, and how their words might affect	
	others in the thread.	
	Use the Context Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context—and vice versa.	
	Judge Behavior, Not Opinions The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.	
	Remain Consistent Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.	
	Output Respond whether the comment you are reading (TGT) is a comment made by a moderator/facilitator on a scale of 1-5, where 1 is "Im sure it was not made by a moderator/facilitator", and 5 is "I'm sure it was made by a moderator/facilitator." Answer with a single number, and nothing else.	
	E.2.4 Single-comment facilitation detection	
	You are an expert moderator/facilitator. You will be presented with a comment (TGT) and the comments made before it (CTX, one for each comment in turn). Choose whether the comment you are reading (TGT) is a comment made by a moderator/facilitator.	
	Key Rules for Annotating:	
	Look at Intent and Impact Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.	
	Use the Context Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context—and vice versa.	
	Judge Behavior, Not Opinions The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.	
	Remain Consistent Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.	
	Output Respond whether the comment you are reading (TGT) is a comment made by a moderator/facilitator on a scale of 1-5, where 1 is "Im sure it was not made by a moderator/facilitator", and 5 is "I'm sure it was made by a moderator/facilitator." Answer with a single number, and nothing else.	